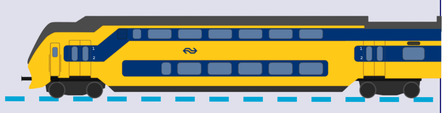
Robust-Rail: An Open-Source Platform for Generating, Solving, and Evaluating the Train Unit Shunting Problem with Service Scheduling

BNAIC 2026

Code:

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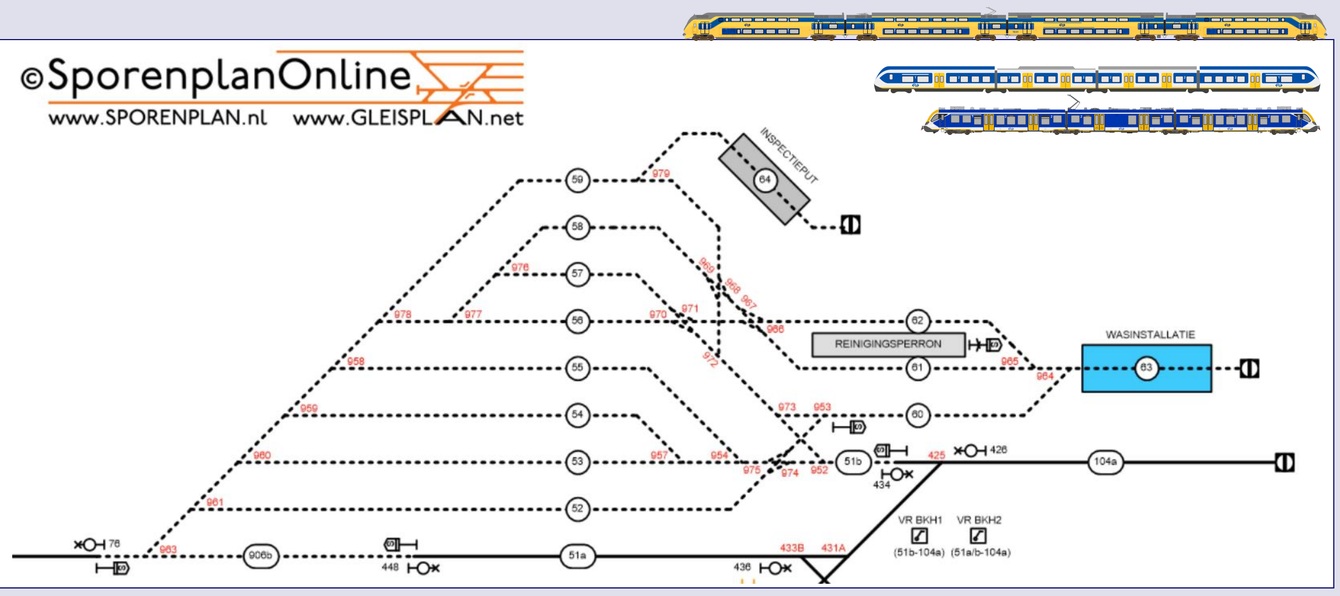
TLDR

An open-source, Docker-based pipeline to generate realistic shunting scenarios, solve them with local search or AI planning (PDDL + ENHSP), and evaluate every plan in one simulator (TORS), all in one shared JSON format.

PROBLEM

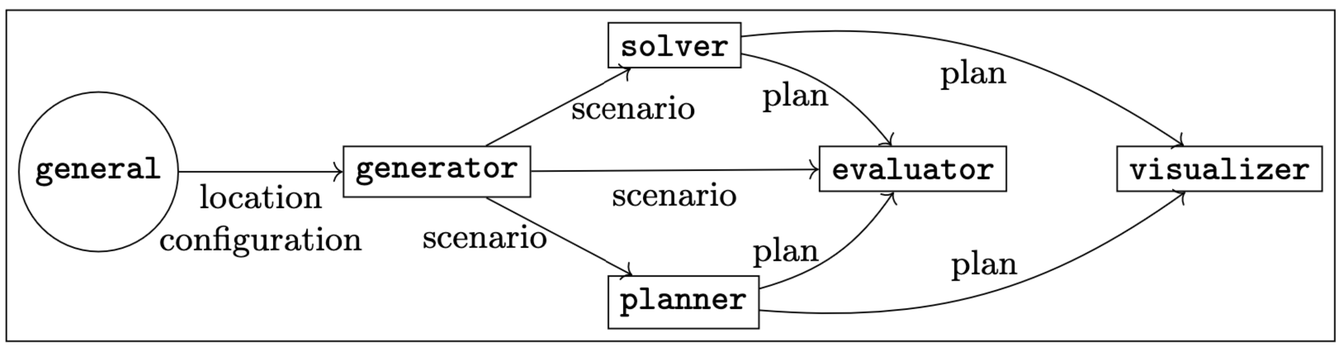
Train Unit Shunting Problem with Service task Scheduling (TUSPwSS)

- **Parking**
- **Routing**
- **Matching**
- **Servicing**



ROBUST-RAIL

general – shared schema, yards & pipeline
generator – random scenarios per yard
solver – HIP: tabu & local search
planner – AI planning: PDDL + ENHSP
evaluator – TORS simulator checks rules
visualizer – animated replay of a plan



https://github.com/Robust-Rail-NL/robust-rail-<module>

PLANNER

PDDL model

- Numeric PDDL, built with Unified Planning
- One driver: one train moves at a time
- Moves follow the A/B side of each track
- (Un)coupling at the front or back of a train
- Matching precompiled into shunting units
- Services only on tracks with that facility
- Goal: every unit parked or departed correctly

From scenario to validated plan

Scenario (JSON) → Precompile matching → PDDL model → **Search ENHSP / h_{rail}** → TORS plan (post-process)

Our heuristic h_{rail}

A weighted sum of progress indicators, so search sees the milestones that h_{add} misses:

unsatisfied goals	unarrived trains	unserviced units	open parking slots
unassembled requests	missing units	pending departures	remaining distance

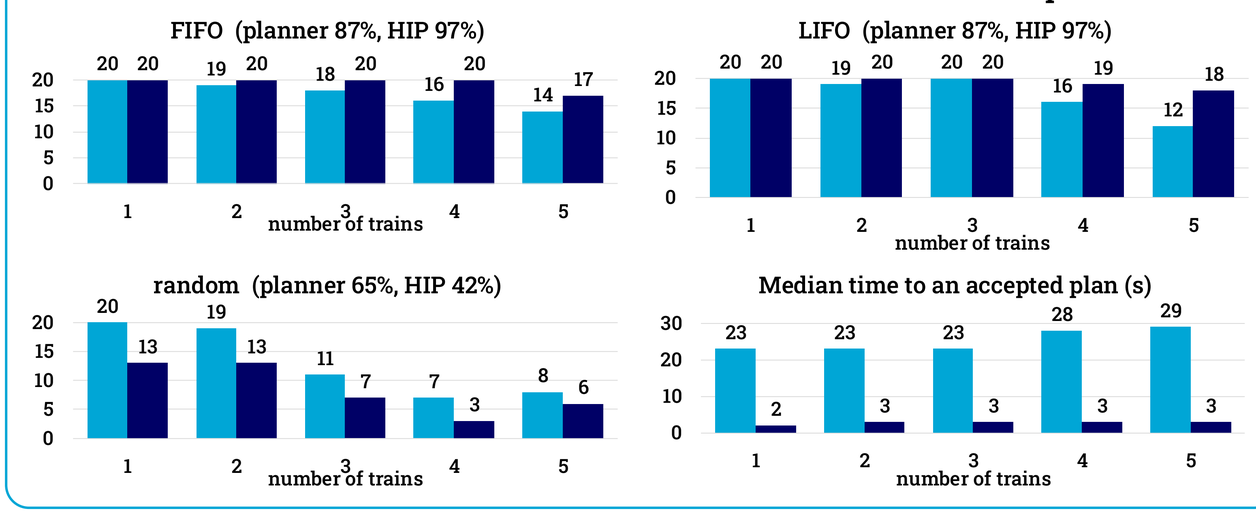
Not admissible: it aims to find a feasible plan fast, not an optimal one.

EXPERIMENTS

Setup

- **Yard: Kleine Binckhorst (The Hague)**
- **Units: VIRM, ICM, SNG, SLT; no servicing**
- **1–5 trains × 3 orders × 20 = 300**
- **Departure order: FIFO / LIFO / random**
- **Planner: PDDL + our heuristic h_{rail}**
- **Baseline: HIP local search**
- **Every plan validated by the TORS evaluator**
- **Time limit: 600 s**

Results Scenarios solved (plan accepted by TORS), out of 20 per bar — planner / HIP



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